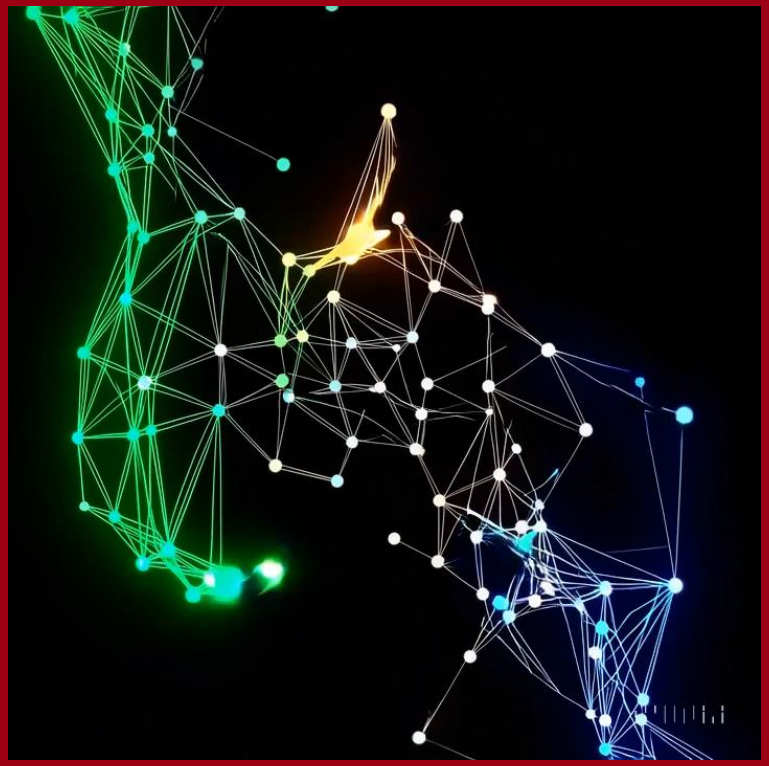
Università degli Studi di Padova

DIPARTIMENTO DI INGEGNERIA DELL'INFORMAZIONE



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Digital Forensics and Biometrics

A.A. 2026/2027

Lecture 13

Social Media / IoT 4N6 / Anomalies


Social Media Forensics

INTRODUCTION

① limited set of data sources in many cases.
② acquiring the server's hard drives is not feasible,
③ leveraging the service operator's data directly requires the service operator's cooperation.

Activities

Acquire and preserve evidences: messahes, post, images, profiles, timestamp

Provenance and integrity: who uploaded this? Was this modified?

Recostruct events: organize posts and interactions into timelines

Analyze relationships and activity: public interactions and account connections for patterns relevant to an investigation

Provide context: Relate content to surrounding posts, media, and events to avoid misunderstanding

Report findings: Clearly separate observed facts from interpretations and uncertainties, and document methods

TRACES

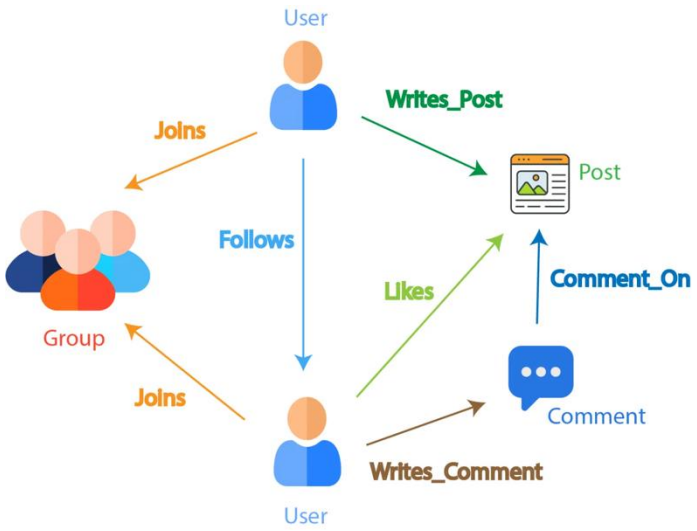
The social footprint: What is the social graph of the user, with whom is he or she connected (friend)?

Communications pattern: How is the network used for communicating, what method is used, and with whom is the user communicating?

Pictures and videos: What pictures and videos were uploaded by the user, on which other peoples pictures is he or she tagged?

Times of activity: When is a specific user connected to the social network, when exactly did a specific activity of interest took place?

Apps: What apps is the user using, what is their purpose, and what information can be inferred in the social context?



SOCIAL FOOTPRINT

- List of friends can be easily retrieved = nodes
- What type of relations exists within a cluster of friends?
  Edges = "connected to", "messaged to" "mentioned in"
- Different data sources can be examined and showed
  - Social Interaction Graph
  - Complete Timeline
  - Localization

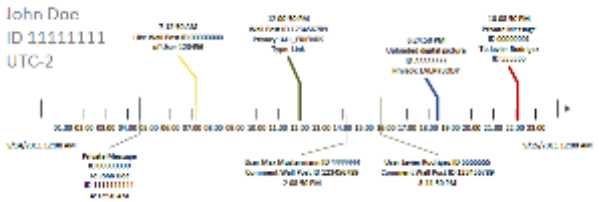
$G = \langle V, E \rangle$ Social graph

$V = \{v_1, v_2, \ldots, v_N\}$ friends

$E = \{e_{i,j} = (v_i, v_j) \ldots\}$ relation

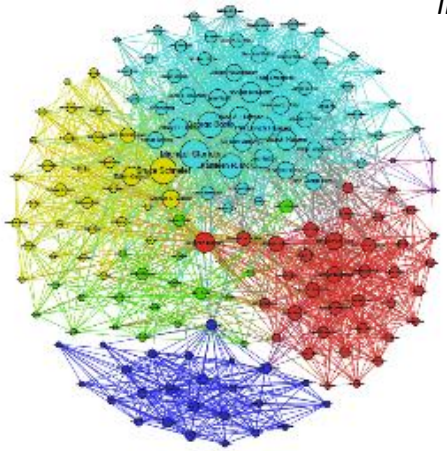
- Available strategies
  - Event tracking (viral scammers, bots, malicious softwares, …)
  - Timeline matching
  - Differential snapshot

Timeline

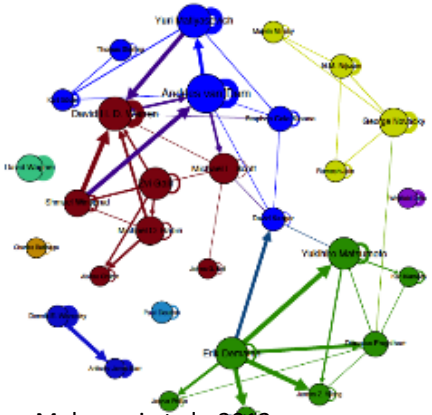


- Data processed in creating social interaction graphs
  - Clustering friends' lists
  - Graph from picture tags
  - Direct messages

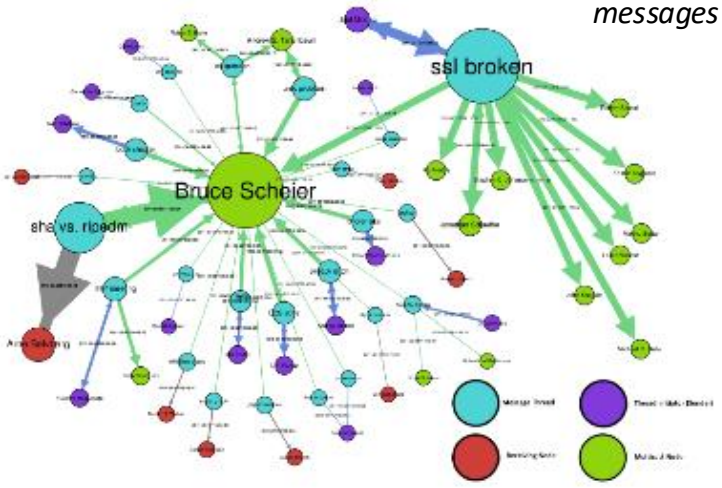
Friends' lists



Picture tags



Direct messages



Images from Mulazzani et al., 2012.

EXAMPLE

Multi-step processing to identify outliers and group member e.g. to identify extremists in a network of supporters

a. Identify a seed list of known extremists; looks for connections and consider the directionality (e.g. TV show)

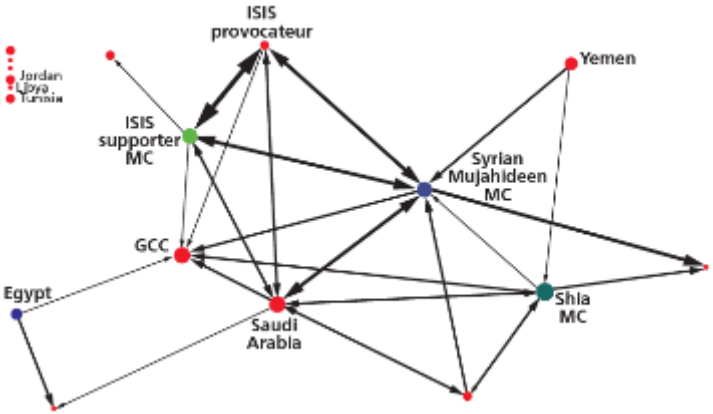
b. Assert cliquishness and in-network focus (e.g. football team supporters) to improve network membership identification

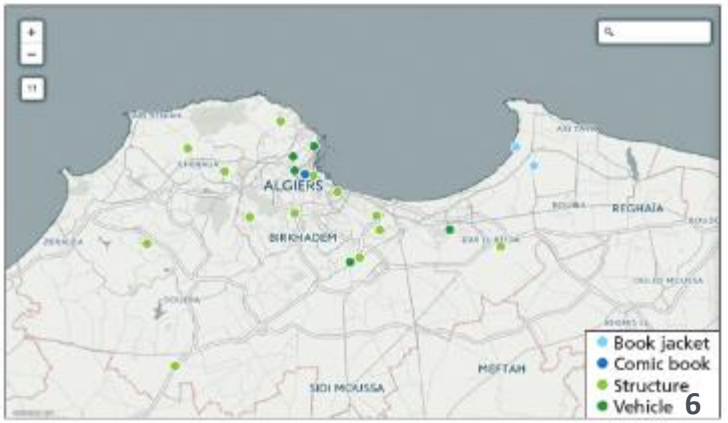
c. Lexical analysis can be used to identify groups or communities
   use of peculiar words or patterns identify communities

d. Resonance analysis: track propagation in time
   i. Build a linguistic model
   ii. Regional matches
   iii. Score users with linguistic model; check whether it happened by chance
   iv. quantify matching (likelihood) in time

e. Stance analysis: detecting messaging strategies

f. Crowdsourcing to understand the information environment
   i. Geotagging, geoinferencing
   ii. Image classification
   iii. Mapping images





Images from Bodine-Baron, Marcellino, et al., 2015.

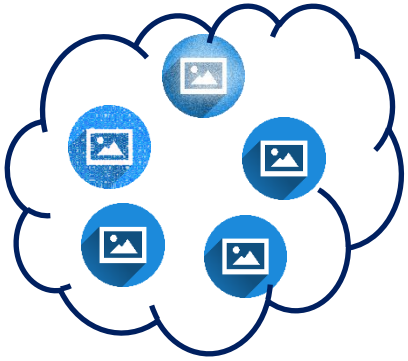
ISSUES

- Data are hardly reproducible: timeline and graphs can look quite different in time
  - data quite big
  - social interactions are dynamic (and therefore, data can be concealed)
- Some data are available only to ISP or SN managers
- How deep must analysis in the interaction graph?
- Sampling bias – users share what they want to share; age; social classes.
- Social media penetration is variable (and depends on SN)
- Legal issues
- Ethics

SPREADING OF DATA THROUGH NETWORKS



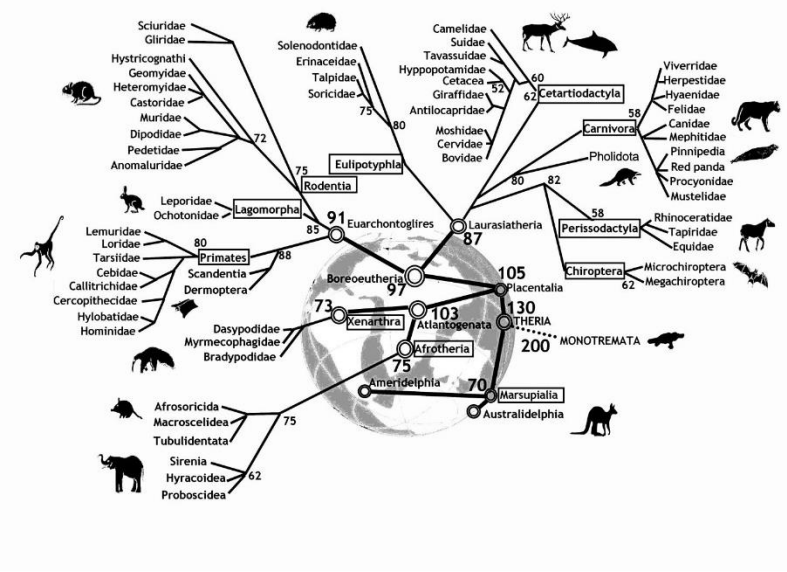
- Social media and data sharing platforms allow pervasive diffusion of digital contents
  - Posts, twits
  - datasets
  - Images and videos
  - software (e.g., github)



- After being available online, contents start spreading
  - Epidemic diffusion models
  - Often contents evolve and change
  - … at the end several copies (or near duplicates) of the same content are available

SOME QUESTIONS ARISE …

Which is the original one? When was it published? What is its origin?



Phylogenetic analysis

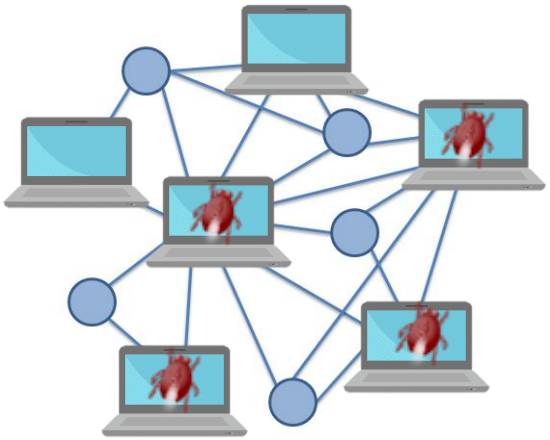
The term was mutuated from biology given the analogy with the analysis of the mutation process that occur to living organisms in biology.

This evolution can be represented by a tree structure named **phylogenetic tree**



It can be done for different media and digital contents.





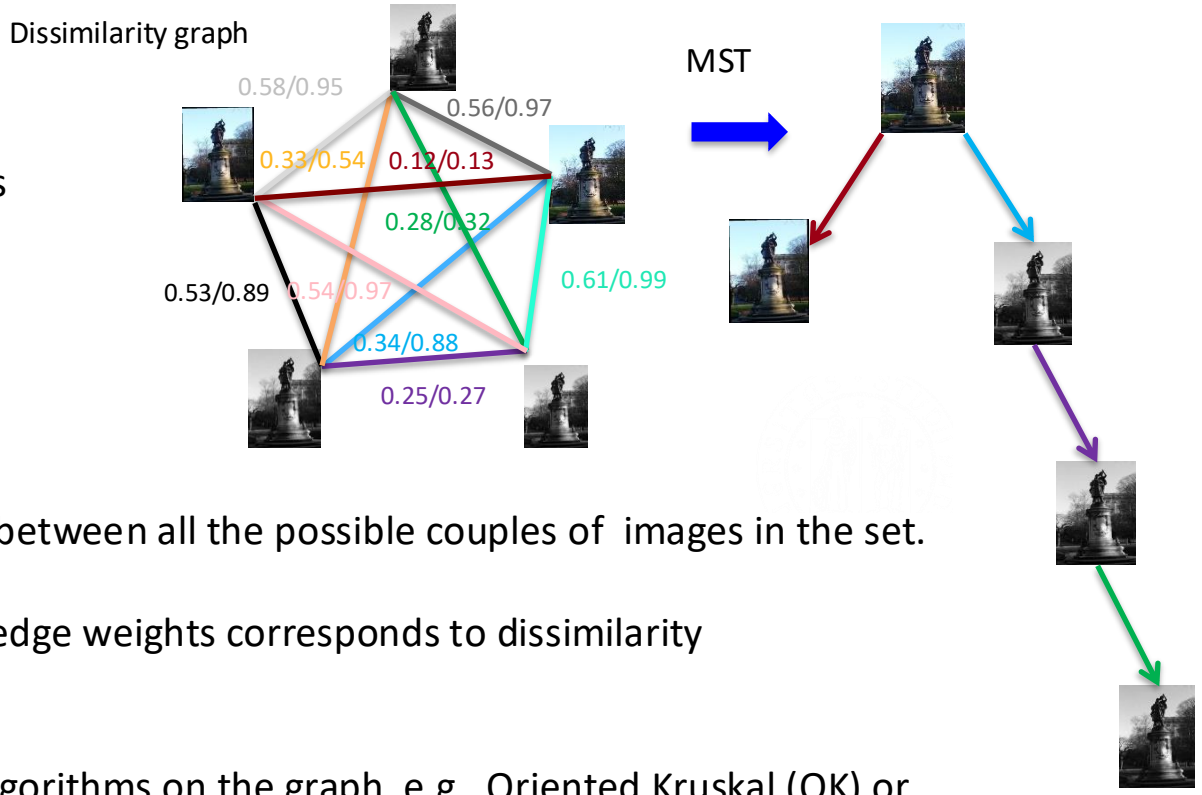
Applications:

- Traceback of illegal contents
- Copyright protection
- Malware profiling
- Fake news detection and debunking

IMAGE PHYLOGENY TREE RECONSTRUCTION

How to reconstruct a phylogeny tree?

Graph-based spanning tree algorithms



① Compute a dissimilarity matrix between all the possible couples of images in the set.

② Build a relational graph where edge weights corresponds to dissimilarity measurements

③ Run **minimum spanning tree** algorithms on the graph, e.g., Oriented Kruskal (OK) or Optimum Branching (OB).

④ Obtain a tree that signals which images are parents and which are children.

Problem: Dissimilarity is an extremely noisy metric!

STATE-OF-THE-ART STRATEGIES

- Improve dissimilarity estimation [Riot *et. al.,* 2017]
- Combine multiple dissimilarity metrics [Costa *et. al.,* 2017, Verde *et. al.,* 2017]
- Probabilistic modelling [Deledalle *et. al.,* 2012]
- Dependency checks [Melloni *et. al.,* 2014]
- Use side information [Milani *et. al.,* 2017]
- Employ new DL-based robust metrics [Yuan *et al.,* 2019]

In our approach, deep learning can be used to denoise dissimilarity and reconstruct the tree at the same time

IoT Forensics

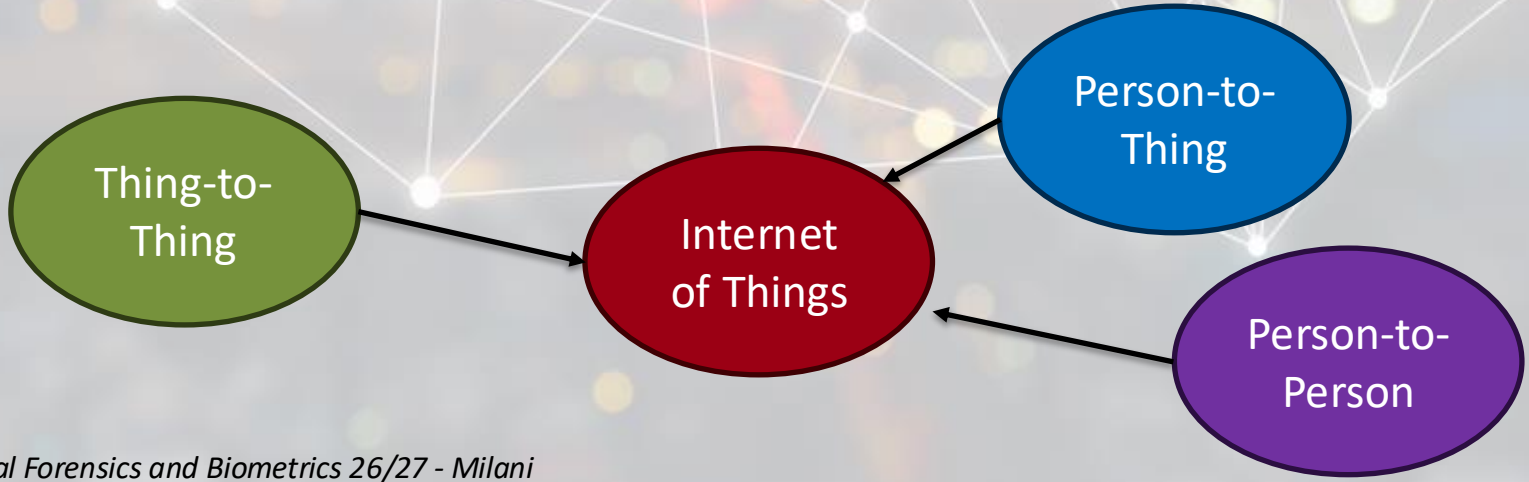


INTERNET-OF-THINGS -> INTERNET-OF-EVERYTHING

Internet-of-Things (IoT) is the interconnection via Internet or other communication networks of heterogeneous devices enabled with sensors, processing ability, softwares.

- Smartphones and related appliances (e.g., speakers)
- Smart homes (e.g., domestic automation)
- Smart cities (e.g., traffic management, infrastructure management)
- Power grids and energy management (e.g., power collection, distribution)
- Manifacturing (e.g., diagnostics, production control, supply chain management)
- Healthcare systems (e.g., smart beds, wristbands, remote monitoring, medical apps)
- Environmental monitoring
- Agriculture

Communication could take place automatically without human intervention



DEVICES

- **Wearable devices:** smart watches, glasses, heart rate sensors, cardioplethysmograph, earbuds, …
- **Smart home appliances**: locks, light sensors, thermal or gas related sensors
- **Control and safety systems for industrial application**: industrial machinery, cameras, failure detectors, fire detectors, logistics automation
- **Vehicles**: Cars, Unmanned Aerial/Ground/Submarine Vehicles (UA/G/SVs),

In 2020, worldwide market was **1.7 trillions $**



NEED FOR IOT FORENSICS

- **Many attack ports/points**
  - -Like having a house with too many windows/doors: extensive attack surface difficult to control/manage
  - -many IoT device are rather "simplistic" (low power/low latency/limited cost): security level is not adequate
  - -device like doors: give malwares access to private networks from outer less-secured spaces
  - -heterogeneity: devices could be very different (security measure must differ as well)

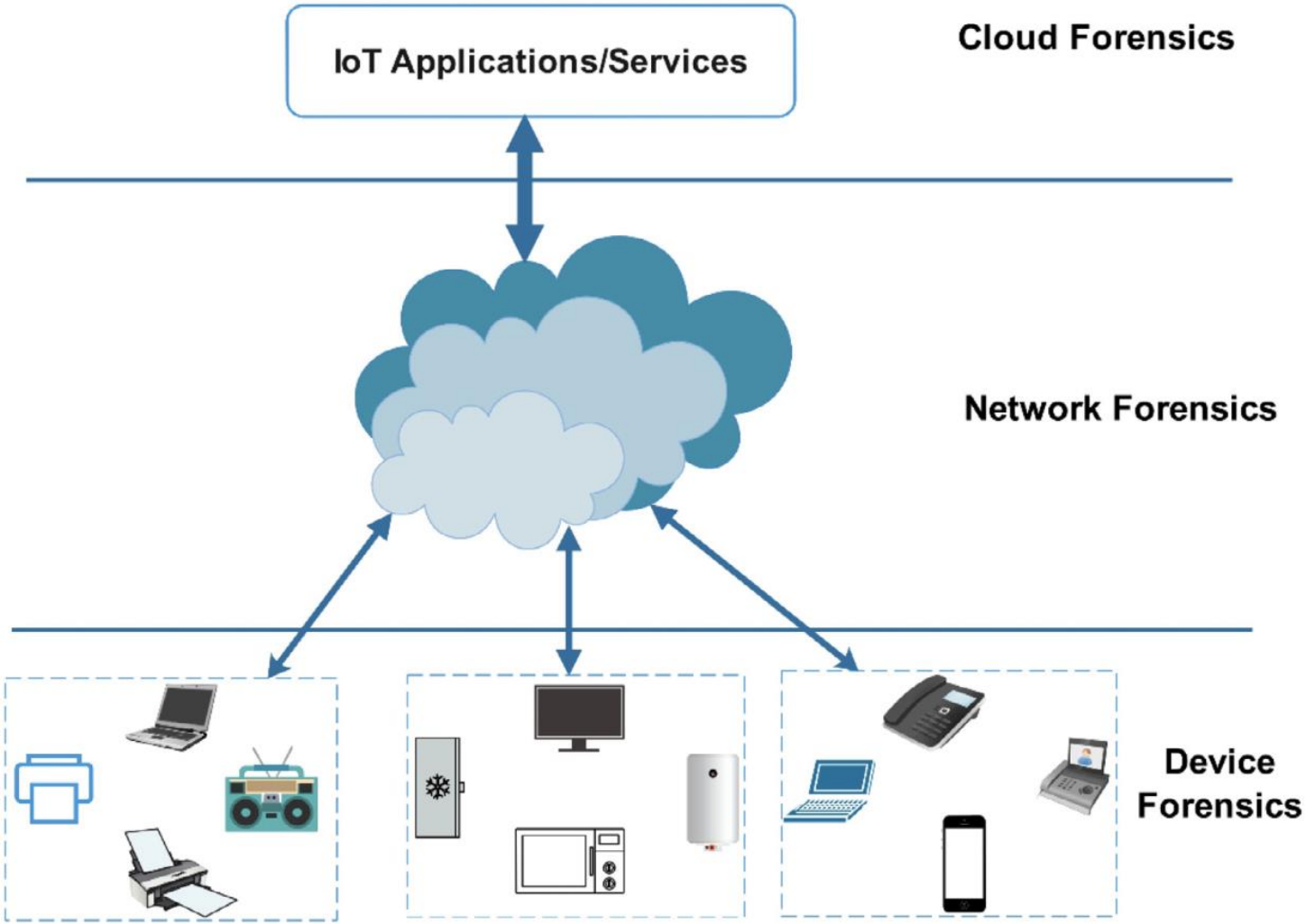
- **New cybersecurity threats**
  - -they could be related to health, privacy and private life, personal goods
  - -they cross boundary of cyberspace and threats human life

  Main issues: **DoS** and **Information leakage**



- **Additional source of information**
  - -can help investigation
  - -can prove or disprove a hypothesis for digital investigators

THREE LEVEL OF ANALYSIS



TRADITIONAL/CLOUD/IOT FORENSICS

Item	Traditional Forensics	Cloud Forensics	IoT Forensics
Number of devices	Billions of devices	Billions of virtual machines and servers and other cloud datacentre components.	It is expected to have 50 billion by 2020 and it is increasing.
Type of Networks	Mobile communication networks, Internet, Bluetooth, Wi-Fi, wired, and wireless networks	Private, public, community and hybrid	RFID and wireless sensor networks (WSNs) such as reader sensor
Protocols	Different types of wireless networks, Ethernet, Bluetooth, IPv4, and IPv6	MTP, EIGRP, AMQP, XMPP, CEE, SSHP, IGMP, SRP, CLNP, and Gossip	RFID and Rime
Network boundaries	Fairly clear boundaries and lines of ownership	Chains of datacentres	Increasingly and there are no defined boundaries
Ownership	Individuals, groups, companies, governments, etc.	Individuals, Cloud Service Providers (CSPs), organizations, governments, etc.	Individuals, groups, companies, governments, etc.
Source of evidence	PC, social networks, authentication, authorization, and accounting servers, mobile devices, and web clients	Virtual machines, virtual servers, network logs, and cloud storage	Home appliances, connected cars, tags, readers, sensors nodes, WSNs, and medical devices.
Type of Evidence	Electronic documents and standard file formats such as JPEG, MP3, etc.	All formats	All formats
Quantity of evidence data	Up to terabytes of data	Petabytes of data	Up to exabytes of data
What to seize	Seize devices as required	Remote access to evidence such as Cloud servers, virtual machines, virtual server (no physical accessibility)	Identify the next best things for the source of evidence

CHALLENGES

General problems

Evidence identification, collection and preservation

Analysis and correlation

Presentation

GENERAL PROBLEMS

Lack of well-defined and tested methodologies

Lack of tools

… which could result in …

Contaminate or destroy evidences

Jeopardize trust and agreements in cross-jurisdictional activities



EVIDENCE IDENTIFICATION, COLLECTION AND PRESERVATION

Detecting presence of IoT system that could be useful in the investigation

Lack of training for DEFR

Wide range of software and hardware specifications

Lifespan limitations



… as a result …

Data can be easily overwritten or lost

Officers could shut down or disconnect the device prior to saving the necessary information

EVIDENCE ANALYSIS AND CORRELATION

Overwhelming amount of data

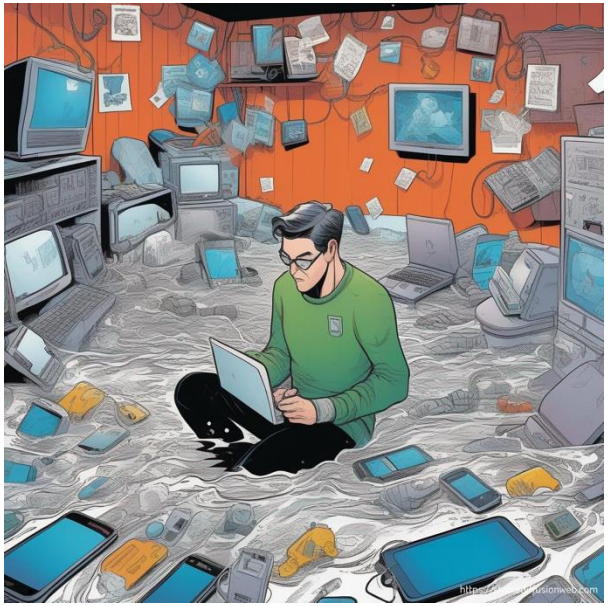
Time lining and limited correlation evidence

No metadata!

..as a result …

Creating a timeline can be diffuclt

Handling huge amount of data can be difficult



EVIDENCE PRESENTATION

Jury has a basic understanding of digital forensics and cloud computing

Explaining technicalities could be very long and challenging

No standardization, so will the methodology be accepted?



… as a result ….

If unusual methodlogy is used, his/her analysis can be significantly challenged in court (risk of removing or altering part of the data must be refuted)

EXAMPLE: ATTACK ON REFRIGERATION 2026

- Hackers have manipulated refrigeration temperatures at supermarkets across Europe, forcing retailers to discard food and exposing security weaknesses in remotely controlled equipment (oct. 2026)
- Attack on blood storage in France (nov. 2026)
- Reset is not enough and possible loss of information



Current operations_
-checking what was compromised and what was not
-removing faulty controllers / isolation of device
-update firmware

In this case safety and continuity take priority over live testing, don't delay for collecting evidences

1. *Stabilize system*
2. *Coordinate IR logging all operations*
3. *Preserve evidence logging records from controllers, … (avoid unnecessary reboots, updates)*
4. *Isolate affected connections with care*
5. *Create timeline and correlation between events*
6. *Assess scope and impact*
7. *Recover systems from trusted sources*
8. *Document and report*

INCIDENT 1: TRENDNET

TRENDnet, Inc. : a producer of wireless cameras (IP cameras), that can be installed wherever you need for a video feed to your electronic devices, such as a computer or a smart phone.

- in 2013, the Federal Trade Commission (FTC) filed a complaint against TRENDnet: they failed to provide reasonable and appropriate security (hacking attacks).

- Hackers posted Internet links to compromised feeds for nearly 700 cameras;
- … displayed private areas of users' homes and allowed the unauthorized surveillance of infants sleeping in their cribs, young children playing, and adults engaging in typical daily activities.

Fiinal settlement: as a first-time offender, TRENDnet was not sanctioned (otherwise, it coud be $16K$ per violation, or more than 11 M$)



https://www.ftc.gov/enforcement/cases-proceedings/122-3090/trendnet-inc-matter

INCIDENT 2: VOLKSWAGEN SCANDAL, 2015

Sept. 18, 2015, the U.S. Environmental Protection Agency (EPA) issued a "Notice of Violation" to the Volkswagen accusing the automaker of cheating on emissions tests (compliancy with Clean Air Act orCAA)

$50,000 grant obtained by the faculty at West Virginia University to evaluate a group of cars (there was no former knowledge of any suspicion)

After all the cars passed their in-house testing, they were taken on the road-test between Los Angeles and San Francisco: different results

After all the cars passed their in-house testing, they were taken on the road-test between Los Angeles and San Francisco and then turned down pollution control: software is protected by the law from being openly reviewed and tested by a third party in the U.S



INCIDENT 3: STRAVA CASE

Strava, a fitness app, visualizes map that shows all the activity tracked by users (people record their exercise and share it with others).

Allows to see popular running routes in major cities, or spot individuals in more remote areas who have unusual exercise patterns: heat maps

weekend military analysts noticed that the map is also detailed enough to give info about a subset of users: US Bases are clearly identifiable and mappable

- Most (all) users in Syria, Afghanistan, Djibuti are US military personnel
- Zooming in on one of the larger bases clearly reveals its internal layout (not visible on Google street view & others)
- Area 51 can be mapped as well



A military base in Helmand Province, Afghanistan with route taken by joggers highlighted by Strava. Photograph: Strava Heatmap

INCIDENT 4: ARKANSAS CASE

James Andrew Bates was charged with omicide: a friends of his was found drowned in his tub (Nov. 2015)

Bates owned an Echo smart speaker. Police investigating the killing have issued a search warrant to Amazon for data from the Echo in Bates's home.

If a smart speaker gets activated prior to or during a crime in the home, it may have recorded sounds, including voices, after it was "summoned." And, the recordings could provide clues as to the perpetrator, timing, victim, and occurrences during the crime.

A device owner could object to the government's effort to access his or her device, arguing that it is an unreasonable search and seizure under the 4th Amendment to the Constitution. The government would have to show probable cause for conducting the search.



CHALLENGES

1. Diversity of IoT environment
2. Complexity of IoT architecture
3. Big IoT data
4. Multi-jurisdiction
5. Privacy concerns
6. Chain of custody and data integrity
7. Security
8. Evidence found through IoT forensics

EXISTING SOLUTIONS

- **Cyber-Trust**
  use Intrusion Detection Systems (IDS) with distributed ledger technology (DLT), for evidence collection and preservation within a smart home environment.

- **Forensics Edge Management System (FEMS)**
  solution for smarthome, combines several security functions; difficult to implement and test

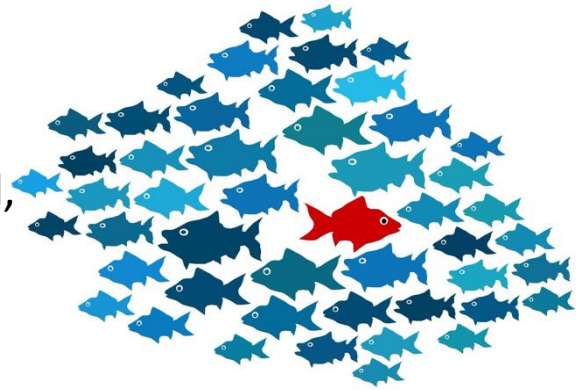
- **PRoFIT Model**
  integrates the (ISO/IEC 29100:2011) standard to ensure and maintain the privacy of witnesses' personal data to encourage voluntary participation

- **Probe-IoT Framework**
  distributed digital ledger to maintain track of all transactions taking place between IoT devices, users, and cloud services

ANOMALY DETECTION

WHAT ARE ANOMALIES?

*"An **anomaly** is an object that is notably different from the majority of the remaining objects."* [1]



Synonims: unusual, irregular, atypical, inconsistent, unexpected, rare, erroneous, faulty, fraudulent, malicious, unnatural, or strange

also called ***outlier*** or ***abnormalities*** vs. ***normal*** or ***nominal*** data

Uses:

- *Fault detection and predictive maintenance*
- Cybersecurity
- Medical analysis
- machine vision
- Statistics
- Neuroscience
- law enforcement
- financial fraud
- …



NOISE ≠ ANOMALY

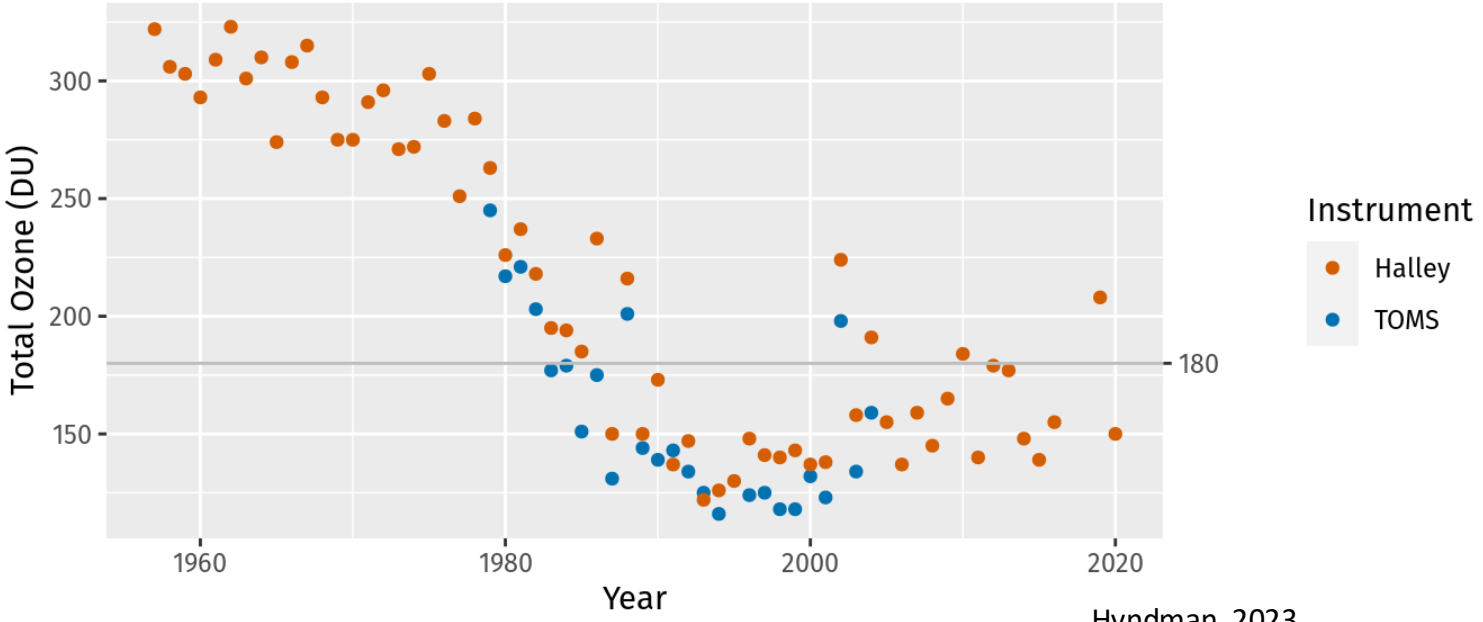
- Noise is erroneous, perhaps random, values or contaminating objects
  Weight recorded incorrectly
  Grapefruit mixed in with the oranges
- Noise doesn't necessarily produce unusual values or objects
- Noise is not interesting
- Anomalies may be interesting if they are not a result of noise
- Noise and anomalies are related but distinct concepts

HYSTORY FACTS: OZONE DEPLETION

In 1985 three researchers (Farman, Gardinar and Shanklin) were puzzled by data gathered by the British Antarctic Survey showing that ozone levels for Antarctica had dropped 10% below normal levels.

Nimbus 7 satellite's instruments did not record similarly low ozone concentrations

The ozone concentrations recorded by the satellite were so low they were being treated as outliers (and discarded)



Hyndman, 2023

URGING ISSUES FOR ANOMALY DETECTION

Key technological factors

- Increased data volume.
  - Complex networks (sensors, IoT, heterogeneous terminals)
  - Lots of user generated data
  - Mobile device with increased data generation capacity
- Data are exchanged and mutate at high speed
- Severe consequences for failure

High failure consequences ➡ *low failure rate*

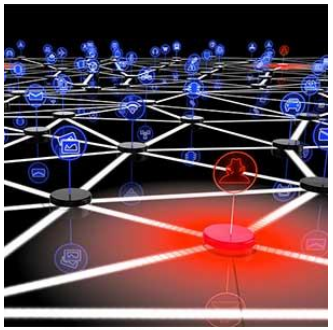
On the detection side:

- Very accurate detectors with machine learning and deep learning strategies
- Data processing capabilities are everyday improving
- Ability to deal with high dimensional spaces and large data sets

Low failure consequences. ➡ *Not-so-low failure rate*

EXAMPLE OF ANOMALIES

- Network attack (anomalous traffic)



- Image tampering or deepfake detection

(local pixel characteristics different from the otherones)

- Fraud detection



- OS attacks and access control analysis
- Malware detection

… but not only for forensics …

- Production chain optimization
- Failure monitoring and prediction
- Predictive maintenance



WHY ANOMALY DETECTION?

- **Unknownness**
  features and characteristics remain unknown until anomalies actually occur

- **Heterogeneous (and evolving) anomaly classes.**
  Anomalies are irregular and changing (adversarial attackes) and thus, one class of anomalies may demonstrate completely different abnormal characteristics

- **Rarity and class imbalance**
  Rare data events; a large amount of labeled abnormal instances are difficult (if not impossible)

- **Diverse types of anomalies**
  Three different types have been explored

- **Need to automatize the process**

THEORY: WHAT IS AN ANOMALY?

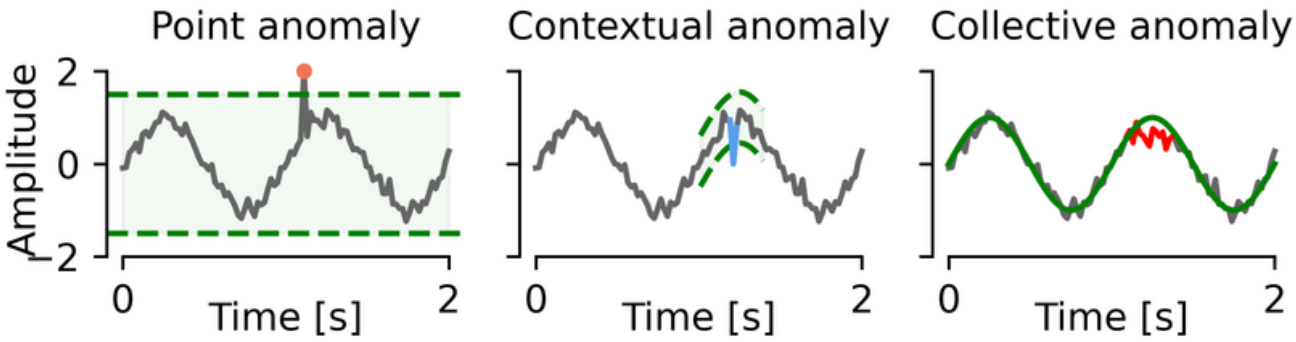
- Define a set of data. $\boldsymbol{x_1}, \boldsymbol{x_2}, \ldots, \boldsymbol{x_n} \in \mathbb{R}^d$
- Mixture of "nominal" samples and "anomalous" samples
- Assume that anomalies are created by a different generation process with respect with the nominal samples

Well-Defined Anomaly Distribution (WDAD) assumption: anomalies are drawn from a known distribution
e.g., repeated instances of machine failures

WDAD is not valid if

- Adversarial conditions (fraud, insider threats, cyber security)
- Diverse set of modes (new failures, not known)
- User's notion of anomaly changes in time (anomaly = interesting point or new data type

TAXONOMY OF ANOMALY DETECTION



Point anomalies

An observation that deviates from trend

e.g. packet losses not expected, credit card transaction

Contextual anomalies

Violation of a seasonal or contextual trend

e.g. 100mm/year rainfall is strange for London, not for Sahara; large credit card expenses on Christmas period

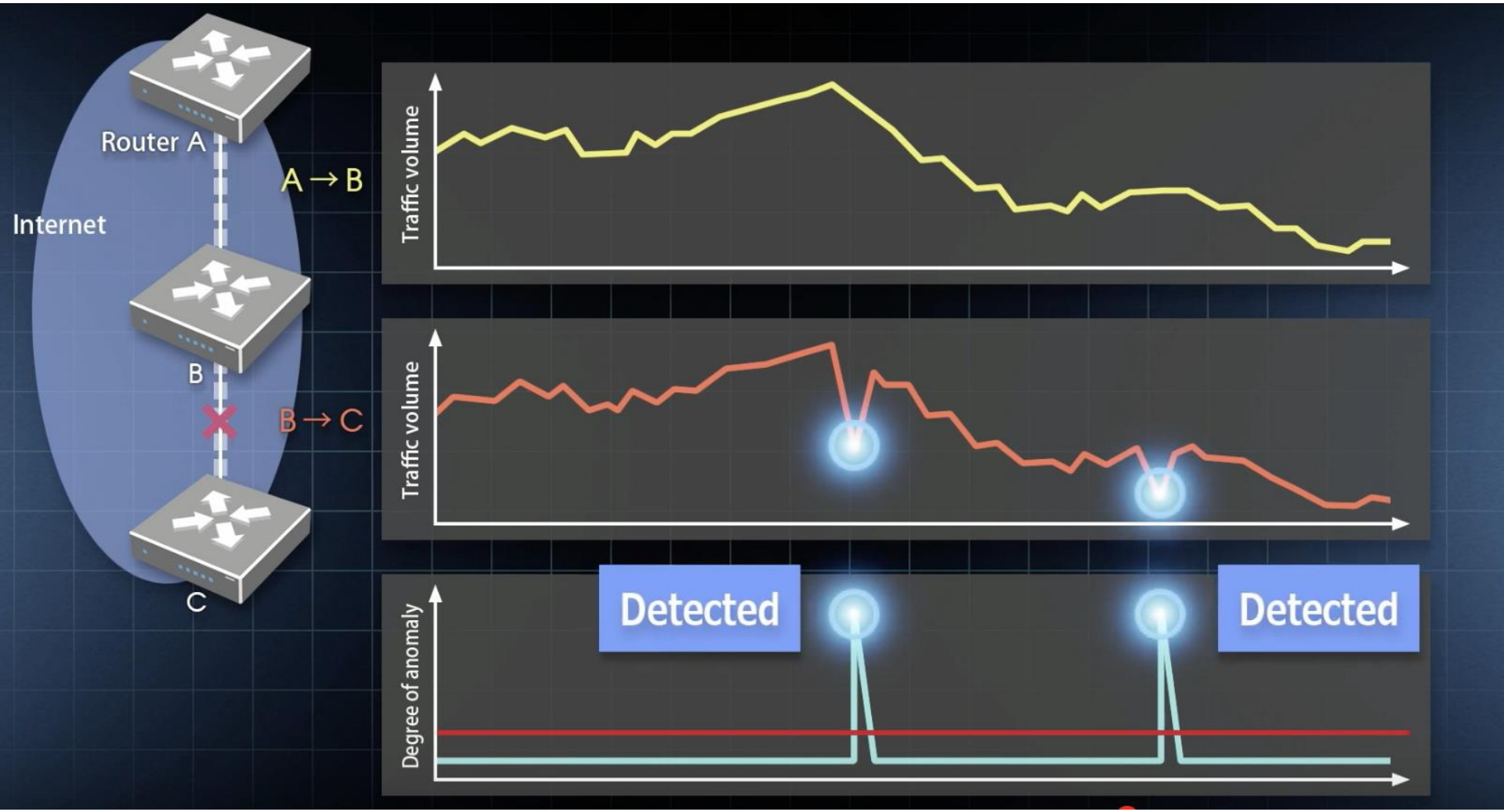
Collective anomalies

Conditional/contextual anomalies w.r.t to the full dataset but not w.r.t. each other (deviation from a pattern)

e.g. heart skip a beat

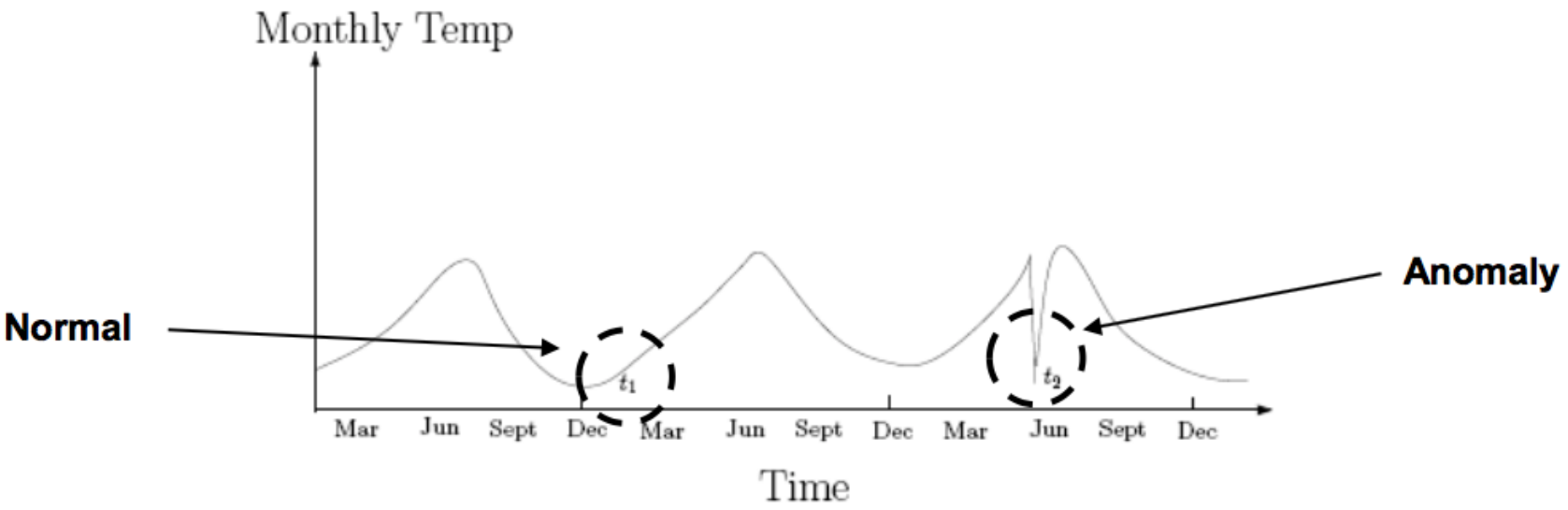
POINT ANOMALIES

Traffic volumes between routers



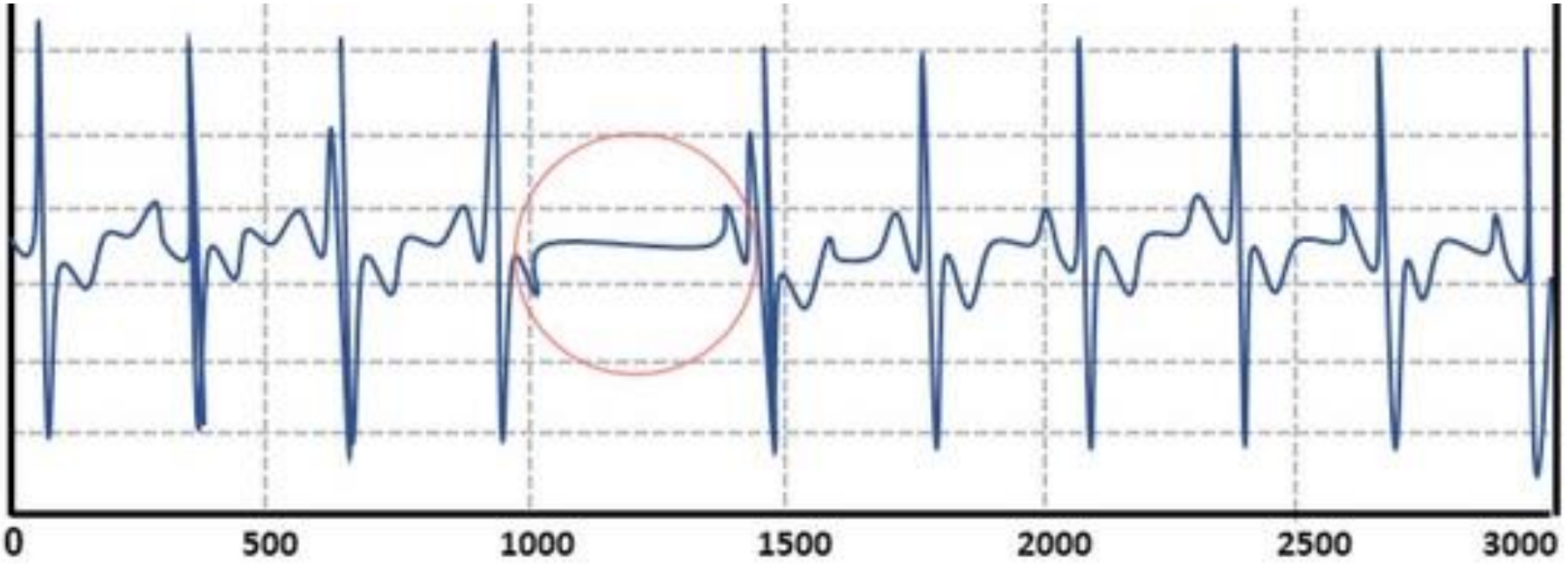
CONTEXTUAL ANOMALIES

Temperature fluctuation per month



COLLECTIVE ANOMALIES

Heart beat



SIGNAL DECOMPOSITION

Signals or time-series (observed) can be decomposed into several parts

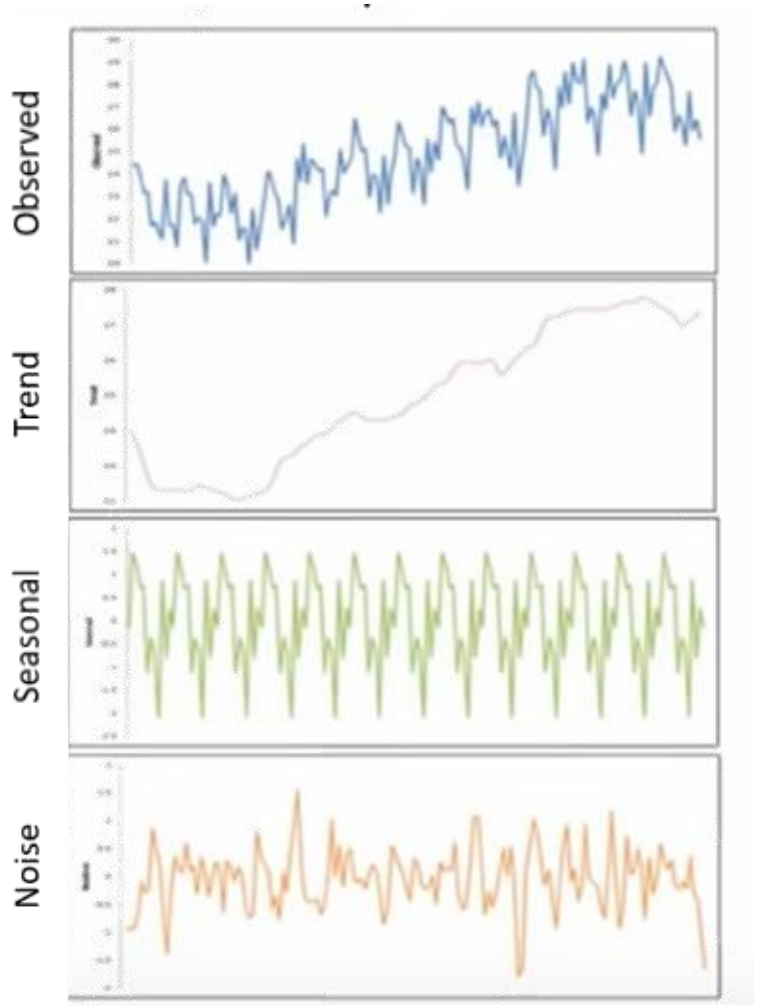
- Trend
- Seasonal components
- Noise

More advanced decomposition:

- Multiplicative decomposition
- Cyclical components

(depends on the signal, e.g., PRNU)

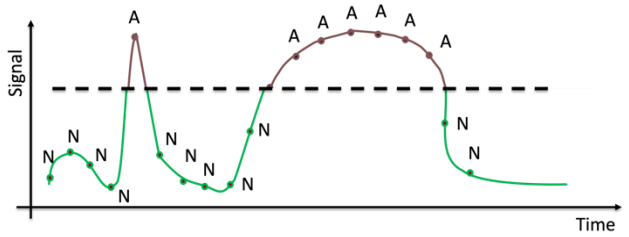
The idea is to extract trends and repetitive components getting rid of noise



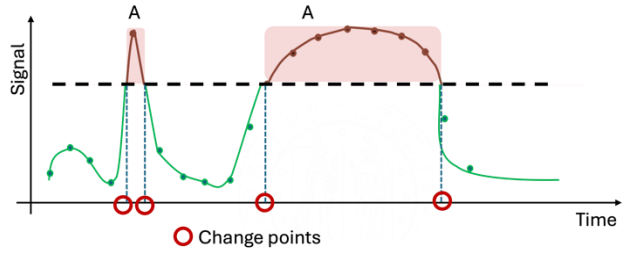
AD EVALUATION

Anomalies can be mainly divided into

- Point anomalies: *outlier detection*

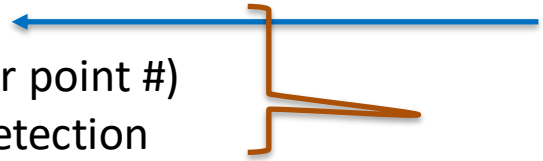


- Collective anomalies (consecutive sequence of anomaly points): *change point detection*



Metrics

- Binary metrics
- Detection time (or point #)
- Window-based detection



outlier detection

change point detection

Binary metrics for change detection applied in window-based way; check if a predicted change point falls into the detection window OR compare between predicted and true windows

AD EVALUATION

- **Binary metrics**

  #occurances with the same label

  Examples: FDR/TPR/Recall/Sensitivity, FAR/FPR, MAR(=1-FDR), Specificity, Precision, Accuracy, G-mean, F-measure, ROC on AUC or PRC, MCC

  Binary metrics for change detection applied in window-based way; check if a predicted change point falls into the detection window OR compare between predicted and true windows

- **Detection time (or point #)**

  differennce between ground truth time and the real one

  Examples: ADD/MAE/Annotation Error, MSD, MSE/RMSE/NRMSE, ADD, Hausdorff

- **Window-based detection**

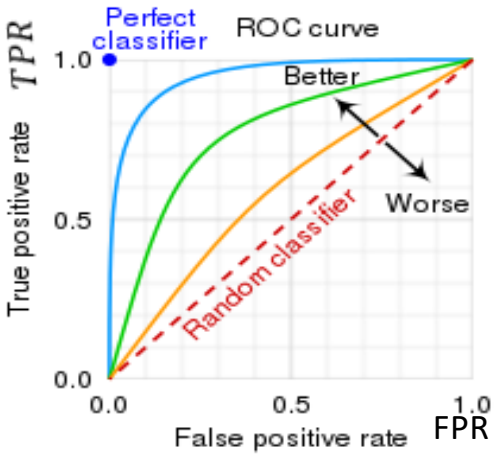
  match predicted change point with the detection window around the actual change point in a way different from binary classification.

  Examples: NAB scoring index, RandIndex

AD BINARY METRICS

Assuming that the decision is binary ("oulier or anomaly" A or "nominal" N) we can have four possible conditions





Confusion matrix

Real \ Estimated	A	N
A	True positive TP	False negative FN
N	False positive FP	True negative TN

- **Precision** $\frac{n_{TP}}{n_{TP}+n_{FP}}$. = $\frac{\text{detected true anomalies}}{\text{samples classified as anomalies}}$ also Positive predicted value (PPV)
- **Recall** $\frac{n_{TP}}{n_{TP}+n_{FN}}$. =. $\frac{\text{detected true anomalies}}{\text{total } anomalies}$. also known as True Positive Rate (TPR) or Sensitivity
- **Specificity** $\frac{n_{FP}}{n_{TN}+n_{FP}}$. =. $\frac{\text{detected false anomalies}}{\text{total } nomimals}$ also known as False Positive Rate (FPR)
- **Accuracy** $\frac{n_{TP}+n_{TN}}{n_{total}}$ accuracy of the classifier
- F1 score $2 \cdot \frac{precision \cdot TPR}{Precision + TPR} = \frac{2TP}{2TP+FP+FN}$. Compares arithmetic and algebraic mean
- AUC. (Area Under Curve). referred to ROC curve

AD BINARY METRICS

- **G-mean** $\sqrt{Sensitivity \cdot Specificity}$

- **MCC** (Matthews correlation coefficient) $\frac{(n_{TP} \cdot n_{TN}) - (n_{FP} \cdot n_{FN})}{\sqrt{(n_{TP}+n_{FP}) \cdot (n_{TP}+n_{FN}) \cdot (n_{TN}+n_{FP}) \cdot (n_{TN}+n_{FN})}}$

measures quality of binary classification

Technique	Recall	FPR	Accuracy	F-1	MCC	Run Time
k-NN	85.71	0.38	97.39	85.71	85.32	≤1
LOF	78.57	0.58	97.38	78.57	77.98	≤1
COF	57.14	1.16	97.35	57.14	55.97	≤1
aLOCI	85.71	0.38	97.39	85.71	85.32	≤69
LoOP	42.85	1.55	97.33	42.85	41.29	≤1
INFLO	57.14	1.16	97.35	57.14	55.97	≤1
CBLOF	92.85	0.19	97.40	92.85	92.66	≤1
LDCOF	85.71	0.38	97.39	85.71	85.32	≤1
CMGOS	57.14	1.16	97.35	57.14	55.97	≤1
HBOS	28.57	1.94	97.32	28.57	26.62	≤1
LIBSVM	85.71	0.38	97.39	85.71	85.32	≤1

In binary metrics, detected change points are labelled as positive if they correspond to ground truth points (*detected* or *missed*).

In anomaly detection, labels are *nominal* or *anomaly*

CHANGE POINT METRICS

- **ADD** (Average Detection Delay) = **MAE** (Mean Absolute Error) = **Annotation Error**
  difference between the detected time and next time for change points

$$\text{MAE} = \frac{\sum_{i=1}^{\#CP} |predicted(CP) - actual(CP)|}{\#CP}$$

- **MSD** (Mean Signed Difference)

$$\text{MSD} = \frac{\sum_{i=1}^{\#CP} predicted(CP) - actual(CP)}{\#CP}$$

- **MSE** (Mean Square Error)

$$\text{MSE} = \frac{\sum_{i=1}^{\#CP} |predicted(CP) - actual(CP)|^2}{\#CP}$$

RMSE = $\sqrt{MSE}$ **NRMSE** = $RMSE / (maxLength(actual(CP)) - minLength(actual(CP))$

- **ADD** at a particular alarm probability

- **Hausdorff**
  Biggest interval between detected CP and its actual value

WINDOW-BASED METRICS

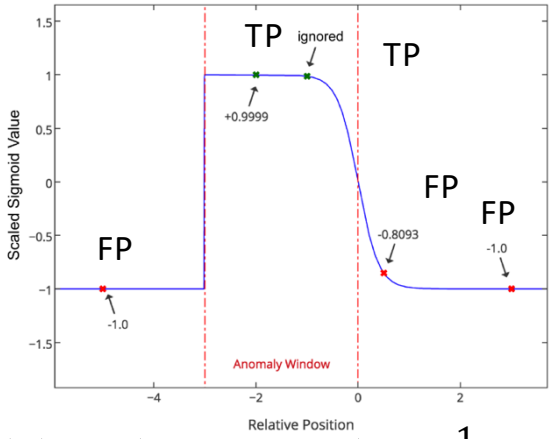
Binary methods do not encorporate time and do not reward early detection: not useful for real-time socring -> window-based scoring,

Evaluate if anomaly falls within a window.

- **NAB** scoring algorithm (Numenta Anomaly Benchmark)
  Used for real-time anomaly detection; rewards early and correct detection and penalizes for false positives and false negatives

- **RandIndex**
  indicates that two (predicted and ground truth) segmentations either agree, or do not agree on pairs of points.

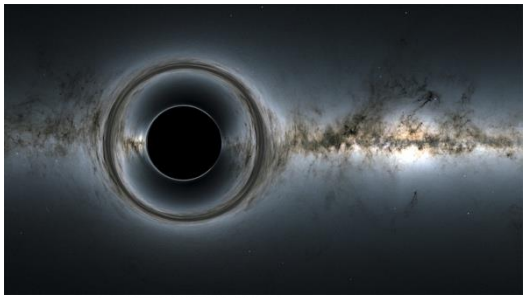


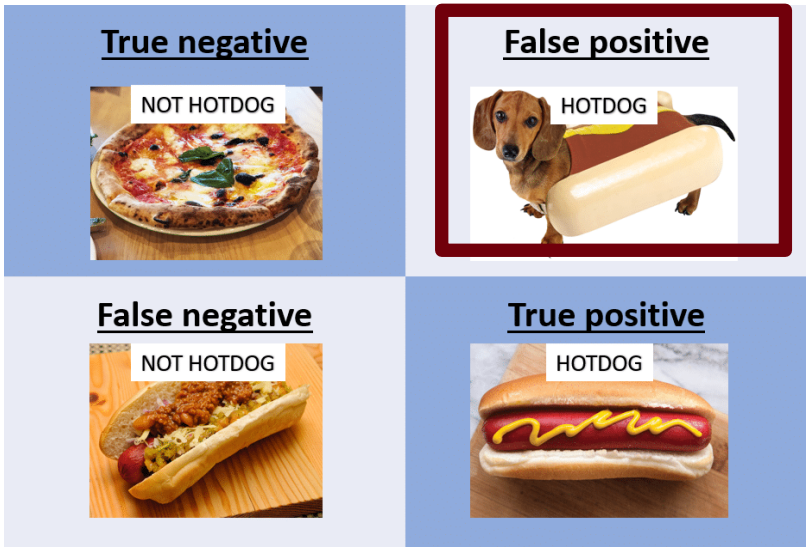


$$\sigma^{A}(y) = (A_{TP} - A_{FP}) \cdot \frac{1}{1+e^{5y}} - 1$$

How to tune the performance?

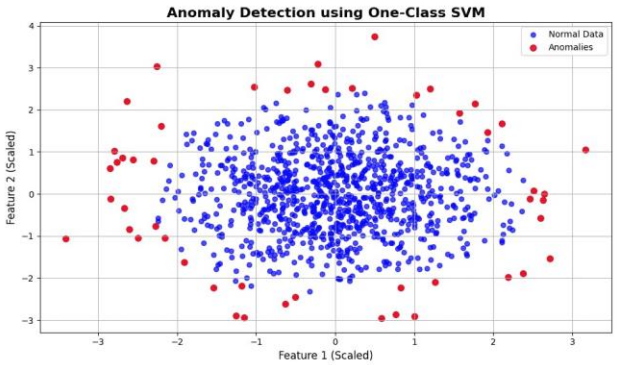
- Automatize choice, but leave some configuration capabilities
- Avoid false alarm black hole
- Allow false negatives, reduce false positive





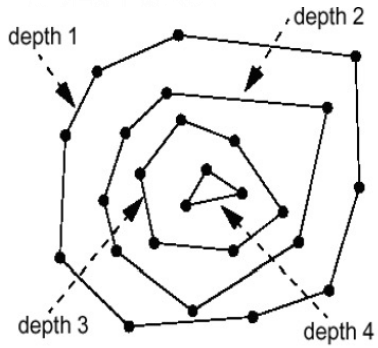
CLASSES FOR CLASSICAL AD ALGORITHMS

- **Statistical or model-based methods**
  normal data follow some statistical model (to be learned); identify the objects in low probability regions



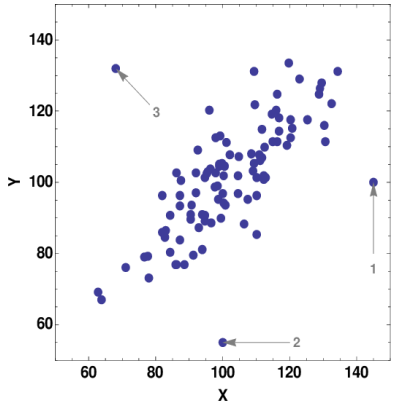
- **Density-based methods**
  object is then considered an outlier if its neighborhood does not have enough other points.

- **Deviation-based**
  Variance change when poonts are removed



- **Neighbor-based or cluster-based methods**
  Anomalies are points far away; different pattern

- **Projection-based methods**
  Project points to reduce dimensionality



- **Bayesian Networks**
  Exploit relational statistics

- **Neural Networks**
  Estimates complex patterns

LEARNING CONDITIONS (1/2)

- **Supervised learning**
  labelled datasets by experts

  Problems: -classes are unbalanced
  -recall is more important than accuracy (catching all anomalies more important than mislabelling nominals)

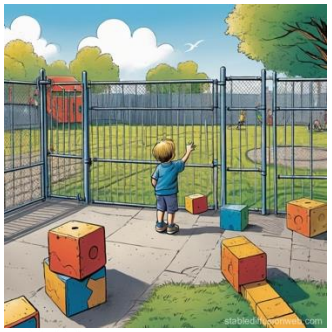


- **Unsupervised learning**
  both training and testing have both classes, no labels. implicit assumption: nominal are somewhat "clustered", anomalies are not.

  Problems:
  -Normal objects may not share any strong patterns, but outliers do in a small area
  -overclustering, high false positive rate and actual outliers undetected.



LEARNING CONDITIONS (2/2)



- **Semi-supervised learning**
  labeled examples and the proximate unlabeled objects to train a model

  - *Clean data**: training set is all made of nominal points, test. set is contaminated with anomalies

  - *Learning from positive and unlabeled examples (LPUE):* two-class problem but only data from one class (the positive class) with unlabeled data points.

  Problems:

  -only some labeled outliers are available, a small number of labeled outliers may not cover the possible outliers well.

MACHINE LEARNING AND DEEP LEARNING

Traditional Learning strategies

- K-means clustering
- Spectral clustering
- Decision trees (Random Forest)
- Support Vector Machine – One-Class SVM
- Self-Organizing Map (SOM)
- Growing Neural Gases (GNG)

Deep Learning

- Fully-Connected networks
- Convolutional Neural Networks (CNN)
- Recurrent Neural Networks (RNN) - Long-Short Term Memory (LSTM)
- Generative Adversarial Networks (GAN)
- One-Shot or Few-Shot Learning architectures
- Transformer networks